The role of mentalizing in pragmatic inference for word retention: an fMRI study

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The sentential context of a novel word–for instance, whether its mapping is ambiguous and must be resolved with pragmatic reasoning–not only impacts initial mapping but longer-term retention (Authors, 2023, 2025a). Behavioral work shows words mapped by resolving pragmatic ambiguity are better remembered than unambiguous meaning-mapping (see Fig 1C for an example of pragmatic inference; here, one must correctly compute speaker's intent and beliefs about their utterance, in absence of clear lexical or syntactic cues, to weigh possible referents against one another). This advantage is related to one's skill at computing others' perspectives, or mentalizing, suggesting there may be greater encoding depth or greater memory accessibility for words learned via mentalizing–a possibility supported by emerging non-linguistic work (Lin et al, 2019; Jimenez & Meyer 2024). However, it's possible that the increased complexity of initial ambiguity resolution is the driving factor. This distinction is critical when considering cross-diagnostic vocabulary growth, with high co-occurrence rates of language and mentalizing difficulties seen in disorders like autism (Durrleman et al, 2015; Tager-Flusberg, 2011; Baron-Cohen et al, 1985; Rosenthal et al, 2019). However, a mentalizing-word retention link hasn't been found in autistic individuals using offline measures (Authors, 2025b). Here, we investigate real-time engagement of the mentalizing network during word learning in non-autistic and autistic adults and how it's linked to the memory system during pragmatic inference.

46 adults (23 autistic (ASD): $M_{Age}=29\pm6$, $IQ_{Verbal}=111\pm19$, $IQ_{NonVerbal}=110\pm11$; 23 non-autistic (NT): $M_{Age}=23\pm6$, $IQ_{Verbal}=109\pm13$, $IQ_{NonVerbal}=108\pm11$) completed an Englich word-learning task during fMRI. Novel words were learned in three conditions–direct mapping (DM, Fig1A), where a clear one-to-one mapping between word and referent exists; lexical inference (LI, Fig 1B), where two novel objects of contrasting sizes are present and initial intended referent ambiguity is resolved by an adjective (e.g., big); pragmatic inference (PI, Fig 1C), where two identical and one unique novel objects are present and intended referent can be resolved by correctly computing the speaker's beliefs/intent. As PI is the only condition where subjects may diverge in their intuitions as to the intended referent, the referent selected by a given subject for a given item in the PI condition would be treated as the answer to be accurately remembered for that subject for retention. Novel word retention was tested 15-20 minutes later (Fig 1E). Subject-specific mentalizing ROIs were captured via a non-verbal localizer (Richardson et al, 2018).

Behaviorally, NT and ASD individuals showed similar initial mapping within each condition (PI: $M_{NT}=0.64\pm0.41$, $M_{ASD}=0.75\pm0.39$; DM: $M_{NT}=1.00\pm0.01$, $M_{ASD}=0.99\pm0.01$; LI: $M_{NT}=0.99\pm0.0.01$, $M_{ASD}=0.99\pm0.01$; Group Difference: *p*s=n.s). For retention, both groups remembered more words in the PI and LI conditions, compared to DM ($p_{vsDM}<0.05$; Fig 2), without significant difference between each other. Across both groups, mentalizing regions were more activated in PI over DM or LI (PI>DM: DMPFC, $p<0.05$; PI>LI: DMPFC, RTPJ, *p*s<0.05; Fig 3) and for PI above baseline (DMPFC, LTPJ, RTPJ; *p*s<0.05; see Fig 1D for example of baseline). There were no significant group differences. However, the activation magnitudes in DMPFC and LTPJ were more positively correlated with those in bilateral hippocampi for NT individuals than for ASD individuals in the PI condition (NT: *r*s>0.35, *p*s<0.01; ASD: *p*s=n.s.; Group Difference: $p_{DMPFC}=0.06$, $p_{LTPJ}<0.01$; Fig 4), suggesting the memory encoding path for pragmatically inferred words was different between the two groups. Groups did not differ for LI and DM, with both having LTPJ-hippocampi and DMPFC-hippocampi activation correlations for LI but not DM (LI: *r*s>0.37, *p*s<0.01; DM: *p*s=n.s.).

Overall, we find evidence that resolution of initial sentential ambiguity, even with limited engagement of the mentalizing network as in lexical inferences, supports novel word retention in both non-autistic and autistic individuals. However, our findings do not contraindicate a possible link between mentalizing and memory, as both pragmatic and lexical inferences, with their high

rates of retention, show a correlation between degree of mentalizing and memory activation. Notably, this link is more widespread in non-autistic individuals. Thus, while neuroimaging concurrent state highlights behaviorally-unattested roles of mentalizing for pragmatic inference resolution in novel word mapping for autistic individuals and reinforces it for non-autistic, further work must tease apart possible distinctions in relation to longer-term retention.

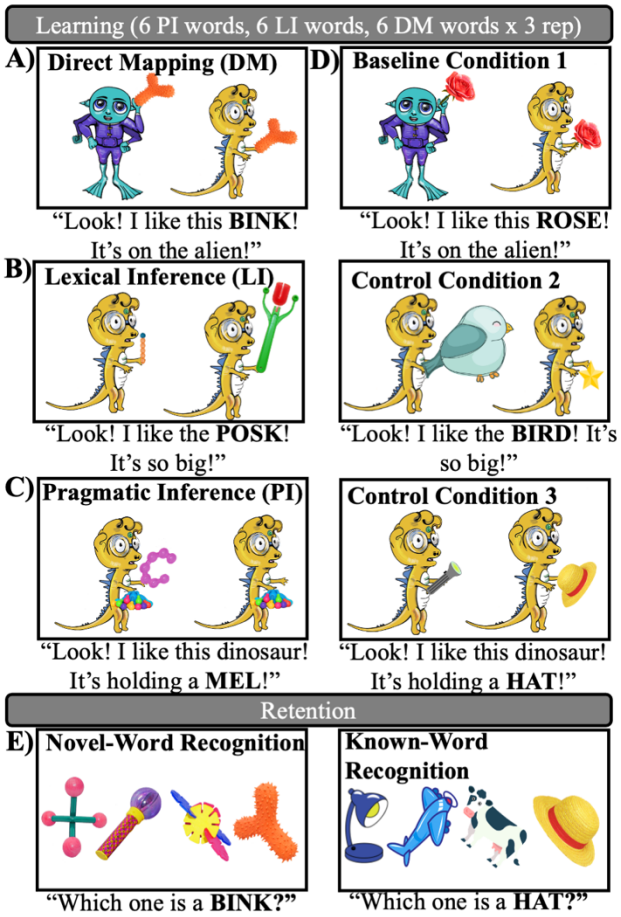

Fig. 1: Experimental stimuli example trials for word learning and retention. A-C represent novel word learning trials, D the corresponding known word trials which serve as syntactically-matched baselines for each novel word condition, and E retention trials. Only novel word trails contribute to behavioral scores. Text in quotations would be spoken aloud.

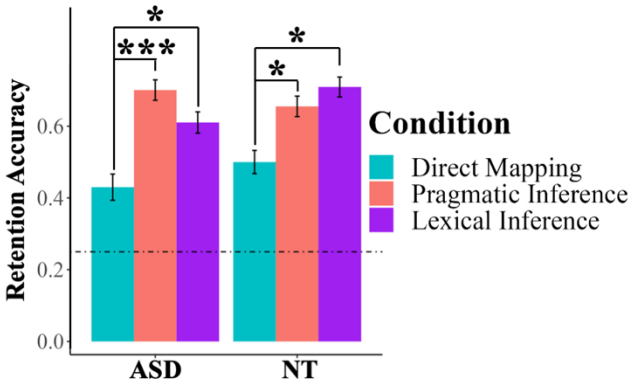

Fig. 2: Mean retention accuracy for novel words for autistic (ASD) and non-autistic (NT) participant by condition. Error bars represent standard error. Dashed line represents chance. *: p<0.05; ***: p<0.001

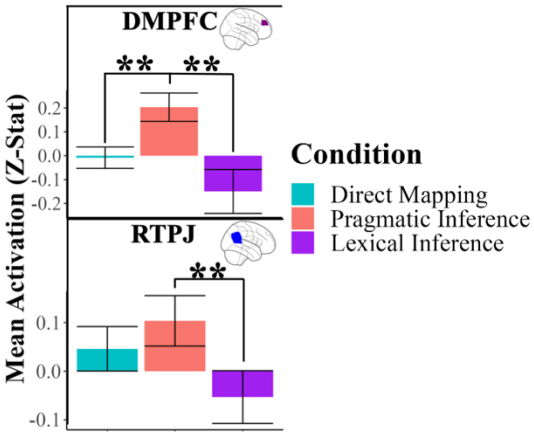

Fig. 3: Mean activation during novel word mapping in mentalizing parcels with between-condition differences. Activation collapsed across group. Error bars represent standard error. **: p<0.01.

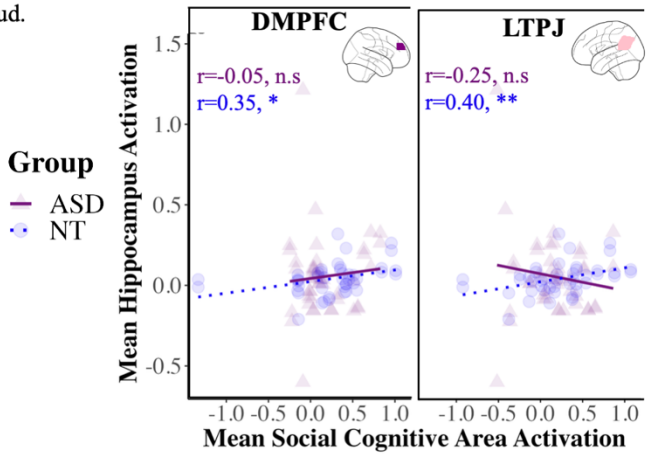

Fig. 4: Mentalizing area-Hippocampal activation correlation for pragmatic inferences for DMPFC and LTPJ. *: p<0.05, **: p<0.01